

# YekBûn — Device Control Mobile APIs (Final)

Complete handoff · Auth not required · Base: <https://api.appdash.yekbun.org/api> · Jul 2026 · Cache v1.15.0

## 1. Boot flow (must follow)

- 1) App cold start → **GET /api/app/device-profile** with hardware params
- 2) Apply **data.runtime** (video/reels/feed/api/rendering/network) + **data.cache** locally
- 3) Heartbeat periodically → **POST /api/app/device-telemetry**
- 4) On crash/ANR → **POST /api/app/device-crash**

There is **no separate cache endpoint** — cache pack is inside resolve.

## 2. Endpoints summary

| Method   | Path                      | Purpose                       | Writes DB?                               |
|----------|---------------------------|-------------------------------|------------------------------------------|
| GET/POST | /api/app/device-profile   | Resolve device+runtime+cache  | No                                       |
| POST     | /api/app/device-telemetry | Heartbeat upsert by device_id | Yes → device_telemetry                   |
| POST     | /api/app/device-crash     | Crash/ANR report              | Yes → problem_devices (+ telemetry bump) |

## 3. Resolve Profile — START API

### GET or POST /api/app/device-profile

Params: **ram** ("4 GB") or **ram\_class** (4|6|8|12+), **cpu\_tier** (entry|low|mid|high|flagship), **os**, **os\_version**, **manufacturer**, **model**, **platform** (optional).

```
GET /api/app/device-profile?ram=4%20GB&cpu_tier=low&os=Android&os_version=13&manufacturer=Samsung&model=SM-A145F
```

Response (Low tier trimmed)

```
{
  "success": true,
  "message": "Device profile resolved.",
  "data": {
    "matched_by": { "ram_class": "4", "cpu_tier": "low", "platform": "android" },
    "profile": {
      "key": "low", "name": "Low", "version": "v1.14.1",
      "cache_profile_key": "low", "runtime_profile_key": "low",
      "memory": { "max": 900, "warn": 700, "critical": 820 },
      "fallback": { "auto_downgrade": true, "max_crash_rate": 2 }
    },
    "runtime": {
      "key": "low",
      "api": { "max_parallel": 3, "retry": "smart", "timeout_ms": 15000, "prefetch": true },
      "feed": { "batch_size": 8, "preload": 2, "strategy": "lazy", "memory_budget": 192 },
      "video": {
        "autoplay": true, "quality": "480p", "max_active": 2,
        "preload": 4, "buffer": 8, "decoder": "hardware", "memory_budget": 384
      },
      "reels": { "initial": 1, "next_preload": 1, "video_queue": 4, "strategy": "lazy" },
      "rendering": { "image_quality": "medium", "animation": "reduced", "strategy": "minimal" },
      "network": { "mode": "conservative", "bandwidth_saver": true, "weak_network_mode": true }
    },
    "cache": {
      "key": "low", "version": "v1.15.0",
      "allocation": { "total_size": 266, "max_size": 319, "mode": "hybrid", "expandable": true },
      "categories": [
        { "id": "feed", "max_size": 32, "priority": "high", "auto_cleanup": true },
        { "id": "video", "max_size": 48, "priority": "high", "auto_cleanup": true },
        { "id": "reels", "max_size": 36, "priority": "high", "auto_cleanup": true }
      ],
      "cleanup": { "mode": "automatic", "order": "priority",
        "protected_categories": ["system", "fonts", "languages", "policy"] },
      "sync": { "on_app_start": true, "wifi_only": false, "mobile_data": true, "max_sync_size": 106 }
    }
  }
}
```

## 4. What mobile MUST apply from runtime

| Field                                      | Meaning                                                 |
|--------------------------------------------|---------------------------------------------------------|
| video.max_active                           | Max videos playing at once (Entry=1, Low=2, Balanced=3) |
| video.quality / preload / buffer           | Quality + preload/buffer seconds                        |
| video.autoplay                             | Entry=false; Low+=true                                  |
| reels.initial / next_preload / video_queue | First load + next prepare + queue                       |
| feed.batch_size / strategy                 | Feed paging; prefer lazy on Entry/Low                   |
| api.max_parallel / timeout_ms              | Network concurrency + timeout                           |

|                             |                                   |
|-----------------------------|-----------------------------------|
| cache.categories[].max_size | Disk cache caps per category (MB) |
|-----------------------------|-----------------------------------|

### 5. Cache budgets by RAM (v1.15.0)

| Tier     | RAM     | Total   | Video | Reels | Feed | Notes                    |
|----------|---------|---------|-------|-------|------|--------------------------|
| Entry    | ≤4 GB   | 169 MB  | 28    | 20    | 20   | wifi-only sync; maps off |
| Low      | 4 GB    | 266 MB  | 48    | 36    | 32   | safe feed scroll         |
| Balanced | 6–8 GB  | 476 MB  | 96    | 64    | 48   | default                  |
| High     | 8–12 GB | 784 MB  | 180   | 120   | 80   | hybrid cleanup           |
| Ultra    | 12 GB+  | 1248 MB | 320   | 200   | 120  | prefetch-friendly        |

### 6. Telemetry Heartbeat

**POST /api/app/device-telemetry · Required: device\_id**

```
{
  "device_id": "D-abc123",
  "user_id": "optional_if_logged_in",
  "name": "Galaxy A14", "model": "SM-A145F", "manufacturer": "Samsung",
  "os": "Android", "os_version": "13", "ram": "4 GB", "cpu_tier": "Low",
  "profile_key": "low",
  "cache_used_pct": 84, "memory_usage_pct": 92, "fps": 32,
  "health_score": 42, "crash": "None", "crash_count": 0,
  "status": "healthy", "app_version": "3.14.2", "app_version_bucket": "latest"
}
```

Upserts by device\_id. Send every few minutes / on app resume.

### 7. Crash / ANR Report

**POST /api/app/device-crash · Required: device\_id, problem\_type**

```
{
  "device_id": "D-abc123",
  "problem_type": "OutOfMemory",
  "crash_cause": "OutOfMemory in FeedRenderer.mountItem()",
  "crash_signature": "OOM · FeedRenderer#mountItem",
  "affected_screen": "Feed",
  "profile_key": "low",
  "os": "Android", "os_version": "13", "ram": "4 GB", "cpu_tier": "Low",
  "manufacturer": "Samsung", "model": "SM-A145F",
  "app_version": "3.14.2", "memory_at_crash": 96, "severity": "Critical"
}
```

problem\_type examples: OutOfMemory, ANR, VideoDecoder, SlowStartup, NativeCrash, JavaScriptCrash, CacheCorruption, LowFPS, NetworkTimeout.

### 8. Mongo collections (QA)

| API                      | Collection                                             |
|--------------------------|--------------------------------------------------------|
| device-profile (read)    | device_profiles + runtime_profiles + cache_profiles    |
| device-telemetry (write) | device_telemetry                                       |
| device-crash (write)     | problem_devices (+ bumps device_telemetry.crash_count) |

### 9. Notes

- Auth not required for these 3 endpoints.
- Content-Type: application/json for POST.
- Seeded tiers: Entry / Low / Balanced / High / Ultra.
- If resolve 404 → backend: php artisan device-control:seed-defaults
- Prefer Entry/Low lazy strategies — never OOM on feed scroll / reels swipe.